

# Margarine and Shortening Effects in Bari Scoring

## Executive summary

“Margarine” is not a nutritionally coherent category anymore. The strongest adverse signal belongs to **industrially produced trans fat** and **partially hydrogenated oils**, not to the word *margarine* itself. WHO identifies partial hydrogenation as the main industrial source of trans fat, reports that PHO commonly contains roughly **25–45% trans fat**, recommends keeping total trans fat intake below **1% of energy**, and estimates that trans fat causes about **500,000 premature cardiovascular deaths each year**. EFSA has likewise concluded that trans-fat intake should be **as low as possible**, and that trans fat may be **more harmful to heart health than saturated fat at equivalent intakes**. <sup>1</sup>

For Bari, the most important design principle is this: **do not penalize all margarine equally**. That would be wrong in both directions. It would under-penalize legacy PHO shortenings and over-penalize modern nonhydrogenated soft spreads. Post-PHO U.S. marketplace analyses found that leading **tub/squeeze spreads** and **stick margarines** had **no or negligible trans fat**, while tub products had materially **less saturated fat** and **more PUFA** than butter and than harder stick products. By contrast, legacy hydrogenated shortenings and older hard-stick products often contained **double-digit trans-fat percentages**. <sup>2</sup>

The health evidence is most decisive for **cardiovascular risk**. Higher trans-fat intake is associated with higher **all-cause mortality**, **CHD mortality**, and **total CHD**, and industrial trans fat has a clearer CHD signal than ruminant trans fat in cohort meta-analyses. Replacing butter or saturated-fat-rich fats with **cis-unsaturated fats**, especially **PUFA**, improves LDL-related risk markers and reduces cardiovascular risk in the broader evidence base. Direct feeding trials found lower LDL when margarine replaced butter, but **PUFA-rich margarine outperformed trans-fat-containing margarine**, which is exactly why Bari should score the underlying fat profile, not the category label. <sup>3</sup>

For **metabolic and inflammation** outcomes, the signal is more layered. Industrial trans fat is linked to inflammatory biomarkers and endothelial dysfunction; replacing SFA with PUFA improves several glucose-homeostasis markers; but randomized meta-analysis did **not** find a consistent overall effect of trans fat on glucose homeostasis alone. Also, the claim that omega-6-rich plant oils are intrinsically pro-inflammatory is **not supported** by randomized-trial meta-analyses. Bari should therefore weight **industrial trans fat** and **saturated-fat-heavy structuring systems** more heavily than generalized “seed oil” or generic emulsifier concerns. <sup>4</sup>

The best Bari implementation is a **split model**. Butter usually scores worse on **fat quality** because of saturated fat, but better on **processing** than a typical retail spread. A modern soft canola/olive/soy spread often scores better on **nutrient profile** but worse on **ultra-processing**. A PHO shortening loses on **both** dimensions and should receive the strongest penalty. Palm-heavy or interesterified trans-free shortenings are intermediate: better than PHO, often worse than liquid-oil-based spreads, and usually still weaker than unsaturated oils on LDL-related risk. <sup>5</sup>

## Health effects and comparative interpretation

The first analytic mistake to avoid is treating butter, margarine, and shortening as simple opposites. They are not. **Butter** is a relatively simple dairy fat that is high in saturated fat and contains naturally occurring ruminant trans fat. **Legacy margarine/shortening** often relied on **partial hydrogenation**, which created industrial trans fat. **Modern soft spreads** are usually built from liquid plant oils with less solid hardstock and now often contain negligible trans fat. **Modern shortenings** are frequently made with palm fractions, fully hydrogenated hardstocks, and/or interesterification to obtain solid functionality without PHO. Bari should therefore score the **fat technology** and **ingredient architecture**, not the label word alone. <sup>6</sup>

The most robust adverse evidence concerns **industrial trans fat**. WHO identifies partial hydrogenation as the industrial source of trans fat, and the European Commission notes that PHOs can contain trans fats from **a few percent to more than 50%**, depending on production technology. In meta-analysis, higher total trans-fat intake is associated with higher risk of all-cause mortality, CHD mortality, and total CHD, while industrial-TFA-specific analyses show a positive relationship with CHD. This is the clearest basis for a major Bari deduction. <sup>7</sup>

Butter, by contrast, mainly loses on **saturated fat** and LDL effects, not on industrial trans fat. In direct feeding trials, replacing butter with margarine lowered LDL by roughly **5–11%**, depending on formulation; PUFA-rich margarine performed better than TFA-containing margarine. Independent trials also found butter raised total and LDL cholesterol compared with olive oil. Broader synthesis from randomized and prospective evidence supports replacing saturated fat with unsaturated fat, especially PUFA, to reduce cardiovascular risk. Bari should therefore rate butter's **processing simplicity** more favorably than retail margarine, while still applying a meaningful **saturated-fat penalty**. <sup>8</sup>

Modern nonhydrogenated soft spreads are a different class from older margarine. In the post-FDA-PHO-ban U.S. market, tub/squeeze products had lower saturated fat and higher PUFA than stick products, and both categories had no or negligible trans fat in the sample analyzed. That means a modern soft spread can reasonably outscore butter on **fat quality**, even if it remains a processed or ultra-processed product. Bari should therefore avoid a blanket “margarine penalty” and instead reward spreads whose dominant oils are **canola, rapeseed, soybean, sunflower, or olive**, with **no PHO** and **low saturated fat**. <sup>9</sup>

Palm-based and interesterified shortenings are intermediate cases. Palm oil and palm-based solid fats are attractive replacements for PHO because they are low in trans fat, but palm oil raises LDL compared with oils lower in saturated fat; in meta-analysis, palm oil increased LDL relative to low-saturated unsaturated vegetable oils. Interesterification changes **triacylglycerol structure** rather than total fatty-acid composition and is widely used to replace trans fat. The human evidence on interesterified fats is not decisively adverse: reviews emphasize uncertainty, and recent controlled trials of commercially relevant palmitic-acid-rich IE fats did not find major worsening in total:HDL ratio or other leading cardiometabolic markers over six weeks. Bari should therefore give IE fats a **moderate caution penalty**, not a PHO-level penalty. <sup>10</sup>

Inflammation and metabolic outcomes are important, but they should be secondary in Bari weighting. Controlled-feeding and observational studies link trans fat to higher inflammatory markers such as CRP, IL-6, and endothelial dysfunction markers. On the metabolic side, replacing SFA with PUFA improves glucose, HbA1c, C-peptide, and HOMA in randomized feeding meta-analysis. Yet the direct RCT meta-analysis of trans fat and glucose homeostasis did **not** find a consistent overall effect on circulating glucose

or insulin. That pattern argues for a strong **CVD/trans-fat penalty**, a moderate **saturated-fat quality penalty**, and a smaller, more cautious metabolic adjustment. <sup>11</sup>

A second common mistake is to treat omega-6-rich plant oils as inherently inflammatory. Systematic reviews and meta-analyses do **not** support that claim. Increasing dietary linoleic acid does not significantly raise inflammatory markers in healthy people, and plant-derived PUFA tends to improve—not worsen—glucose-insulin homeostasis when replacing SFA. Bari should therefore **not** penalize soybean, sunflower, canola, or corn oil merely because they are “seed oils.” The relevant negative signals are **PHO, industrial trans fat, excess saturated structuring fat**, and **ultra-processing**, not the seed-oil name itself. <sup>12</sup>

Sterol/stanol-fortified spreads are a genuine exception worth separating. EFSA and meta-analyses support LDL lowering from plant sterols/stanols, with the effect increasing up to around **3 g/day** and averaging about **10–12% LDL reduction** in that dose range. The Bari implication is narrow: a sterol-fortified spread may deserve a **small positive adjustment**, but only if a realistic serving delivers an evidence-based dose and the product is not simultaneously weakened by high sodium, high saturated fat, or poor overall formulation. <sup>13</sup>

## Comparative synthesis

| Archetype                               | Best-supported health interpretation                                                                                                         | Bari implication                                                                                                | Source                                                                     |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Legacy PHO margarine or shortening      | Strongest adverse profile because industrial trans fat increases CHD risk and mortality; inflammatory effects are also documented.           | Highest penalty; ingredient-triggered red flag.                                                                 | WHO, BMJ meta-analysis, industrial-TFA cohort meta-analysis. <sup>14</sup> |
| Butter                                  | No industrial trans-fat issue, but typically very high in saturated fat; raises LDL relative to olive oil and many unsaturated alternatives. | Penalize saturated fat materially, but do not assign an industrial-trans penalty.                               | Direct trials and broader replacement evidence. <sup>15</sup>              |
| Modern soft plant-oil spread            | Usually negligible trans fat and lower SFA than butter; better lipid profile than older harder products.                                     | Often highest-scoring margarine-type product on nutrient quality, but still usually loses points on processing. | Post-PHO marketplace analyses. <sup>16</sup>                               |
| Palm-heavy trans-free shortening/spread | Better than PHO on trans; usually worse than unsaturated oils on LDL because of saturated fat load.                                          | Moderate penalty focused on saturated-fat-heavy structuring system, not a blanket palm ban.                     | Palm meta-analyses and shortening composition studies. <sup>17</sup>       |

| Archetype                      | Best-supported health interpretation                                            | Bari implication                                               | Source                                    |
|--------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------|
| Sterol/stanol-fortified spread | Can lower LDL at meaningful daily dose; benefit is dose- and serving-dependent. | Small conditional bonus only if effective intake is practical. | EFSA and LDL meta-analysis. <sup>18</sup> |

## Food composition and formulation

The composition question matters because Bari scores products, not slogans. The same broad product class can be built by very different technologies: **partial hydrogenation**, **full hydrogenation plus blending**, **interesterification**, or **palm-based fractionation**. Those choices strongly change trans fat, saturated fat, and the MUFA/PUFA balance. Full hydrogenation largely removes trans fat but creates a very hard saturated fat, which is then usually blended or interesterified to become usable in shortening or stick margarine. Interesterification changes TAG structure and functionality without necessarily changing the overall fatty-acid distribution. <sup>19</sup>

### Typical composition bands for common margarines and shortenings

| Product archetype                      | Typical structuring approach                                                                         | Trans fat % of total fat | Saturated % | Monounsaturated % | Polyunsaturated % | Bari reading                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------|-------------|-------------------|-------------------|-----------------------------------------------------------------------------------|
| Butter comparator                      | Milk fat                                                                                             | ~2–4                     | ~52–60      | ~20–26            | ~2–4              | High saturated-fat burden; no industrial-trans flag unless mixed with other fats. |
| Modern tub/squeeze margarine or spread | Liquid plant oils + emulsifiers + modest hardstock                                                   | Negligible               | ~29         | ~36               | ~33               | Best fat profile within margarine-type spreads in current U.S. sample.            |
| Modern stick margarine                 | More solid structuring fat; often palm fractions and/or fully hydrogenated/interesterified hardstock | Negligible               | ~38         | ~34               | ~29               | Harder texture generally costs more saturated fat than tub products.              |

| Product archetype                                    | Typical structuring approach                 | Trans fat % of total fat                                                  | Saturated % | Monounsaturated % | Polyunsaturated % | Bari reading                                                                   |
|------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------|-------------|-------------------|-------------------|--------------------------------------------------------------------------------|
| Legacy hydrogenated shortening                       | Partial hydrogenation                        | ~12–19 in analyzed commercial shortenings; PHO ingredient averages ~25–45 | ~24–33      | ~28–40            | ~22–26            | Worst formulation class for Bari because trans is intrinsic to the technology. |
| Palm-based shortening                                | Palm oil/palm stearin without PHO            | ~0.1–0.8                                                                  | ~46–52      | ~30–37            | ~7–10             | Major improvement over PHO on trans, but still saturation-heavy.               |
| Interesterified palm-olein shortening                | Palm-based hardstock + IE                    | ~0                                                                        | ~45         | ~43               | ~11               | Trans-free, but still an engineered solid-fat system; moderate caution.        |
| Interesterified palm stearin + cottonseed shortening | IE blend of hardstock + more unsaturated oil | ~0                                                                        | ~43         | ~27               | ~26               | Better PUFA balance than palm-only systems, but still highly engineered.       |

These bands should be treated as **formulation archetypes**, not immutable constants. Different studies report either percent of total fat or percent of total fatty acids, and retail products vary by market and year. But the directional picture is stable: **PHO shorteners are trans-heavy; palm-based and IE systems are trans-light but often saturation-heavy; modern soft spreads are usually the best margarine-type products on fat quality.** <sup>25</sup>

The composition drift over time is large enough that older observational studies on “margarine” need careful interpretation. Garsetti’s U.S. marketplace analysis noted that older USDA reference values for stick and tub products were around **20%** and **7% trans fat** of total fatty acids, respectively, whereas later marketplace analyses found current leading products with **no or negligible trans fat**. Bari should therefore treat older cohort findings about “margarine” as partly historical evidence about **old formulations**, not today’s category as a whole. <sup>26</sup>

## Regulation and labeling

Regulation matters because Bari ingests label data. A scientifically correct engine can still be operationally wrong if it assumes the trans-fat line is always present or always truthful on a per-100 g basis. In practice, the **EU and Israel may not require trans-fat disclosure on the nutrition panel for EU-style labels**, while the **U.S. requires trans-fat declaration but allows rounding to 0 g when a serving contains less than 0.5 g**. Bari therefore needs **ingredient-list parsing**, **per-100 g normalization**, and **source-aware handling** of missing trans values. <sup>27</sup>

### Regulatory limits and labeling rules relevant to Bari

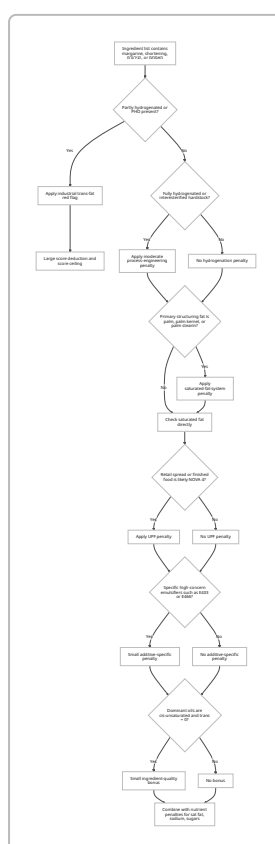
| Jurisdiction   | Core trans-fat rule                                                                                                                                                                                                                           | Labeling rule most relevant to Bari                                                                                                                                                                                                                    | Important blind spot                                                   | Bari consequence                                                                                           | Source                                      |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| European Union | Industrial trans fat in food for final consumers and retail must not exceed <b>2 g per 100 g fat</b> ; foods could remain on market until <b>1 April 2021</b> . B2B suppliers must provide information when trans fat exceeds that threshold. | Mandatory nutrition declaration includes fat, saturates, carbohydrate, sugars, protein, and salt; <b>trans fat is not in the mandatory list</b> . Ingredients must specify <b>“fully hydrogenated”</b> or <b>“partly hydrogenated”</b> where relevant. | Missing trans value on the panel does <b>not</b> mean trans is absent. | Parse ingredient list aggressively; use saturated fat and hydrogenation flags when trans grams are absent. | EU 2019/649 and EU 1169/2011. <sup>28</sup> |

| Jurisdiction  | Core trans-fat rule                                                                                                                                                                                                                                                                                                 | Labeling rule most relevant to Bari                                                                                                                                                                                                                                                                                              | Important blind spot                                                                               | Bari consequence                                                                                                                              | Source                                                  |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Israel        | The Ministry of Health states that, from <b>1 January 2025</b> , imported foods compliant with EU Regulation 1169/2011 may be marketed under the updated regime; Ministry materials also describe longstanding nutrition labeling including trans fat, and Israel aligns key food-law implementation with EU rules. | Official public-facing materials still describe nutrition labels as including trans fat, cholesterol, and saturated fat; a 2025 Ministry Q&A says that under the EU-style regime <b>trans fat is not mandatory</b> on the nutrition panel. Israel's front-of-pack red symbols focus on <b>saturated fat, sodium, and sugar</b> . | Transitional inconsistency between older local labeling practice and newer EU-style import regime. | Treat Israeli data as mixed-regime. Do <b>not</b> rely on the trans line being present; do rely on ingredient cues and saturated-fat profile. | Ministry of Health and EfshariBari pages. <sup>29</sup> |
| United States | FDA determined <b>partially hydrogenated oils are no longer GRAS</b> , effectively removing PHOs from most food uses.                                                                                                                                                                                               | Nutrition Facts requires trans fat declaration, but if a serving contains <b>&lt;0.5 g</b> , it may be listed as <b>0 g</b> .                                                                                                                                                                                                    | "0 g trans fat" per serving can still imply nonzero trans fat on a per-100 g basis.                | Convert U.S. labels to per-100 g and always inspect ingredients, especially in older databases and bakery products.                           | FDA labeling and PHO determination. <sup>30</sup>       |

The practical implication is simple: Bari should assign the **highest confidence** to a direct PHO/partly-hydrogenated ingredient flag; **medium confidence** to a declared trans-fat gram value; and **lower confidence** to the absence of a trans line on the panel. That hierarchy follows both the science and the regulations. <sup>31</sup>

## Bari scoring rules and worked examples

The right Bari design is a **feature-first** model, not a category-first model. In plain terms: **“contains margarine” is a weak proxy; “contains PHO” is a strong signal.** The flow below operationalizes the science and the regulatory reality. It gives the heaviest weight to **industrial trans fat**, strong weight to **saturated-fat-heavy solid-fat systems**, smaller but real weight to **UPF/process engineering**, and only limited weight to **specific emulsifier concerns**, because the evidence there is still emerging. <sup>32</sup>



## Processing and ingredient flags that should influence Bari

| Flag                                                       | Suggested<br>Bari treatment                                  | Rationale                                                                                                       | Evidence<br>strength | Source                                               |
|------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------------------|
| <b>Partly/partially hydrogenated oil or fat</b>            | <b>Very large deduction;</b><br>also cap final score ceiling | This is the clearest ingredient-level marker of industrial trans fat, the strongest adverse formulation signal. | Very strong          | WHO, FDA, EU rules, CHD meta-analyses. <sup>33</sup> |
| <b>Declared trans fat &gt; 0</b> in a non-dairy fat system | Large proportional deduction                                 | Direct nutrient evidence supported by mortality/CHD meta-analysis.                                              | Very strong          | BMJ 2015, WHO. <sup>34</sup>                         |



| Flag                                                                             | Suggested Bari treatment                                  | Rationale                                                                                                                                                                             | Evidence strength | Source                                                                    |
|----------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------|
| <b>Fully hydrogenated oil</b>                                                    | <b>Do not</b> auto-fail as trans fat                      | Full hydrogenation is fundamentally different from partial hydrogenation; it is used to create hardstock with negligible trans, but often raises saturation.                          | Strong            | Review and EU ingredient-naming rules. <sup>35</sup>                      |
| <b>Interesterified fat</b>                                                       | Moderate deduction                                        | IE is a trans-replacement technology with mixed-to-reassuring short-term human data but sparse long-term outcome evidence.                                                            | Moderate          | IE briefings/reviews and recent controlled trials. <sup>36</sup>          |
| <b>Palm/palm kernel/palm stearin as primary structuring fat</b>                  | Small-to-moderate deduction tied to saturated-fat context | Palm systems are usually better than PHO on trans, but worse than lower-saturated oils on LDL.                                                                                        | Moderate          | Palm meta-analyses and shortening analyses. <sup>17</sup>                 |
| <b>Retail margarine/ spread likely classed as NOVA 4</b>                         | Moderate processing deduction                             | NOVA commonly uses margarine as an UPF example; overall UPF exposure is associated with worse health outcomes, though much evidence remains observational and low/very-low certainty. | Moderate          | NOVA references and BMJ umbrella review. <sup>37</sup>                    |
| <b>Specific high-concern emulsifiers such as CMC E466 or polysorbate 80 E433</b> | Small additional deduction                                | Human and cohort evidence suggests potential harm, but this is not yet as strong as trans-fat or saturated-fat evidence.                                                              | Emerging          | Gastroenterology RCT and BMJ cohort. <sup>38</sup>                        |
| <b>Lecithin alone or mono-/diglycerides alone</b>                                | No standalone deduction beyond UPF signal                 | Current direct evidence does not justify a large ingredient-specific penalty at Bari scale.                                                                                           | Weak              | By inference from the specific-emulsifier literature above. <sup>39</sup> |

| Flag                                                                | Suggested Bari treatment              | Rationale                                                                                             | Evidence strength       | Source                                              |
|---------------------------------------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------|
| <b>Sterol/stanol fortification at clinically meaningful serving</b> | Small bonus only if dose is realistic | LDL lowering is real, but dose- and serving-dependent and should not erase poor baseline formulation. | Strong for LDL lowering | EFSA and dose-response meta-analysis. <sup>18</sup> |

## Proposed quantitative adjustments

The cleanest production architecture is to keep Bari's generic nutrient-profile score and then apply a **margarine/shortening module** as a targeted modifier. To avoid double counting, Bari should **not** add a heavy standalone "total fat" penalty for products whose category is inherently fat-based, such as spreads or shortenings; for those products, **fat quality** matters more than fat quantity. That is a better scientific fit than penalizing all grams of fat equally. <sup>40</sup>

A workable rule-set is:

- **PHO rule:** if ingredient list contains *partly/partially hydrogenated*, deduct **35 points** and cap final score at **40**.
- **Trans-fat rule:** deduct **12 × trans fat g/100 g**, cap **36 points**. If all trans is clearly from dairy fat and there is no industrial-trans signal, Bari can optionally half-weight or ignore this line item and let saturated fat carry the penalty.
- **Saturated-fat rule:** deduct **0.8 × max(0, saturated fat g/100 g – 4)**, cap **24 points**.
- **Sodium rule:** deduct **0.02 × max(0, sodium mg/100 g – 120)**, cap **10 points**.
- **Added-sugars rule:** deduct **0.5 × max(0, added sugars g/100 g – 5)**, cap **12 points**.
- **Total-fat rule for mixed foods only:** deduct **0.2 × max(0, total fat g/100 g – 17.5)**, cap **6 points**.
- **UPF rule:** deduct **8 points** when the finished retail product is plausibly NOVA 4.
- **Interesterified-fat rule:** deduct **4 points**.
- **Palm-structuring rule:** deduct **4 points** when palm/palm kernel/palm stearin is the obvious primary structuring fat; increase to **6** when saturated fat is extreme.
- **High-concern emulsifier rule:** deduct **2 points each** for E433 or E466, cap **4 points**.
- **Unsaturated-oil bonus:** add **4 points** when cis-unsaturated oils are dominant, there is **no PHO**, trans is **0**, and saturated fat is modest for the category.
- **Sterol/stanol bonus:** add **2 points** only when a realistic serving delivers a clinically relevant dose.

## Worked examples using a 100-point starting score

| Hypothetical product                               | Key assumptions per 100 g                                                                          | Applied rules                                               | Adjustment | Illustrative final score |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------|--------------------------|
| <b>Legacy frozen pie crust with PHO shortening</b> | Total fat 24 g; trans 2.5 g; saturated fat 10 g; sodium 450 mg; added sugars 2 g; PHO present; UPF | PHO –35; trans –30; sat –5; sodium –7; total fat –1; UPF –8 | <b>–86</b> | <b>14</b>                |

| Hypothetical product             | Key assumptions per 100 g                                                                                                         | Applied rules                                            | Adjustment | Illustrative final score |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------|--------------------------|
| <b>Packaged butter croissant</b> | Total fat 22 g; saturated fat 13 g; sodium 350 mg; added sugars 6 g; no PHO; packaged UPF                                         | Sat -7; sodium -5; added sugars -1; total fat -1; UPF -8 | <b>-22</b> | <b>78</b>                |
| <b>Soft canola-olive spread</b>  | Fat-based category; saturated fat 12 g; trans 0 g; sodium 250 mg; no added sugars; no PHO; UPF; dominant oils are cis-unsaturated | Sat -6; sodium -3; UPF -8; unsaturated-oil bonus +4      | <b>-13</b> | <b>87</b>                |

These examples show the decision logic Bari should aim for. A **legacy PHO shortening** should score dramatically worse than either butter or a modern soft spread. A **butter-based product** can outrank a trans-rich shortening because it avoids the industrial-trans red flag, yet still score below a **modern soft unsaturated spread** because of its saturated fat burden. A typical **palm/interesterified cookie or frosting system** would usually fall between the PHO example and the soft spread: much better than PHO, often worse than a truly soft nonhydrogenated spread, and heavily influenced by added sugars and UPF penalties. <sup>41</sup>

## Source hierarchy and research gaps

### Primary and official sources to anchor Bari rules

| Priority | Source                                            | Why it should anchor Bari                                                                                                     | Citation      |
|----------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------|
| Highest  | WHO trans-fat fact sheet and REPLACE policy basis | Defines the industrial source, intake guidance, and public-health urgency.                                                    | <sup>42</sup> |
| Highest  | FDA PHO final determination                       | Makes PHO a direct regulatory red flag for U.S.-market products and datasets.                                                 | <sup>43</sup> |
| Highest  | EU Regulation 2019/649                            | Gives the hard legal limit of <b>2 g trans fat per 100 g fat</b> and B2B disclosure rule.                                     | <sup>44</sup> |
| Highest  | EU Regulation 1169/2011                           | Explains why EU-style labels may omit trans fat from the mandatory panel and why “fully/partly hydrogenated” wording matters. | <sup>45</sup> |
| Highest  | EFSA opinions on trans fat                        | Supports “as low as possible” intake and the judgment that trans fat may be worse than SFA at equivalent intakes.             | <sup>46</sup> |
| High     | de Souza et al., <i>BMJ</i> 2015                  | Best high-level mortality/CHD meta-analysis for trans fat.                                                                    | <sup>47</sup> |

| Priority    | Source                                                | Why it should anchor Bari                                                                               | Citation |
|-------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------|
| High        | Bendsen et al., <i>Eur J Clin Nutr</i> 2011           | Distinguishes industrial from ruminant trans fat in CHD risk.                                           | 48       |
| High        | AHA Presidential Advisory 2017                        | Strong replacement framing for SFA versus unsaturated fats.                                             | 49       |
| High        | Judd 1998; Denke 2000; Tonstad 2001                   | Direct butter-versus-margarine feeding evidence relevant to product scoring.                            | 50       |
| High        | Weber et al. 2021; Garsetti et al. 2016               | Best category-specific evidence showing how margarine composition changed over time.                    | 51       |
| Moderate    | Sun 2015 and Fattore 2014 palm-oil meta-analyses      | Best evidence for palm as a trans-free but saturation-heavy replacement.                                | 52       |
| Moderate    | Berry/Mensink IE reviews and Hall/Ng IE trials        | Best current evidence for interesterified fats: important, but less definitive than trans-fat evidence. | 36       |
| Moderate    | Lane et al., <i>BMJ</i> 2024                          | Best broad synthesis for UPF, with the important caveat that much evidence is low/very-low certainty.   | 53       |
| Emerging    | Chassaing 2022 and Sellem 2023                        | Best current human evidence for additive/emulsifier-specific concern.                                   | 38       |
| Conditional | EFSA plant sterols/stanols + Ras et al. meta-analysis | Supports a small, narrow bonus for clinically effective fortified spreads.                              | 18       |

## Major uncertainties and research gaps

The biggest uncertainty is **historical drift in product composition**. Many older cohort studies that report on “margarine” likely captured exposures from periods when stick margarine and commercial shortening were far richer in trans fat than current reformulated products. A 2025 Framingham analysis linked margarine intake to worse cardiometabolic outcomes, but the likely relevance of older formulations limits how directly that finding transfers to 2026 retail spreads. Bari should treat such studies as **historically informative, not formulation-agnostic truth**. <sup>54</sup>

The second major gap concerns **interesterified fats**. They are clearly important in the modern food supply and technologically central to trans-fat replacement, but long-term hard-outcome evidence remains thin. The short-term trial evidence is not strongly adverse, which means Bari should apply a **moderate penalty with uncertainty**, not treat IE as a PHO-equivalent hazard. <sup>55</sup>

The third gap is **UPF causality and additive mechanisms**. The epidemiologic signal for UPF is broad and consistent, but much of the evidence base remains observational and low/very-low certainty. Emulsifier-specific evidence is promising enough for a small additive-sensitive penalty, but not strong enough for large

score swings relative to trans fat or saturated fat. Bari should therefore rank evidence roughly as: **industrial trans fat > saturated-fat-heavy fat system > UPF classification > specific emulsifiers.** <sup>56</sup>

A fourth uncertainty is **natural versus industrial trans fat**. WHO states that naturally occurring and industrially produced trans fats are both harmful, but cohort meta-analysis has shown a clearer signal for **industrial** trans fat and a less certain one for **ruminant** trans fat, partly because the data are much sparser. Since Bari's specified criteria include the **presence of industrial trans fats**, Bari should keep a distinct **industrial-trans ingredient flag** rather than treating all total-trans grams as equivalent. <sup>57</sup>

Finally, there is a real **data-engineering gap** in Israel and international imports. Under EU-style labels, trans fat may be absent from the mandatory nutrition panel; in the U.S., it may appear as 0 g despite nonzero content below 0.5 g/serving. Bari therefore needs source-aware ingestion rules: **ingredient parsing first, nutrient line second, per-100 g normalization always.** <sup>58</sup>

## Implementation handoff

### Assessment

Bari should not score on the words מרגרינה / מחמאה / **margarine / shortening** alone. It should score on a **fat-technology stack**: industrial trans fat, hydrogenation status, saturated-fat-heavy structuring system, interesterification, UPF status, and only then additive-specific concerns. The scoring module above is ready to implement and back-test.

### Owner

Head of Product

### Why this owner

This is primarily a prioritization and scoring-policy decision: Bari must explicitly choose to separate **nutrient quality** from **processing burden** and avoid a simplistic category penalty.

### Copy-paste prompt

Design a Bari policy update for margarine/shortening handling. Treat category names as weak proxies and score on direct formulation signals instead: PHO/partially hydrogenated oil as the strongest red flag, saturated-fat-heavy structuring systems as the next signal, IE fats as moderate caution, UPF as a secondary penalty, and soft nonhydrogenated unsaturated-oil spreads as potentially favorable on nutrient quality. Produce a one-page product spec with score ceilings, penalty precedence, and edge-case handling for butter, palm-based shortenings, and sterol-fortified spreads.

### Assessment

The nutrient-science logic is strong enough to formalize now. The remaining task is not more open-ended research; it is codifying which features deserve **very strong**, **moderate**, or **emerging** weight.

## Owner

Chief Nutrition Officer

## Why this owner

This owner should lock evidence strength, especially for industrial trans fat, SFA quality, IE fats, and the limited role of omega-6 or emulsifier penalties.

## Copy-paste prompt

Convert the margarine/shortening evidence into Bari nutrition policy. Approve these evidence tiers: very strong for industrial trans fat/PHO, strong for replacing SFA with cis-unsaturated fats, moderate for palm-heavy structuring fats, moderate-uncertain for IE fats, and emerging for specific emulsifiers and UPF-independent mechanisms. Return a signed-off policy note with exact “strong / moderate / emerging” labels and any changes to the proposed point weights.

## Assessment

The main engineering challenge is **label parsing and score precedence**, especially across Israeli, EU-style, and U.S. labels.

## Owner

Frontend Architect

## Why this owner

Implementation requires deterministic ingredient parsing, per-100 g normalization, and explainable score outputs in the product UI.

## Copy-paste prompt

Implement a Bari fat-technology parser for margarine/shortening signals. Required detections: partly/partially hydrogenated, fully hydrogenated, interesterified, palm oil, palm kernel, palm stearin, butter, plant sterols/stanols, E433, and E466; also support Hebrew terms including מרגרינה and מחמאה. Apply the proposed modifier logic after the core nutrient score, normalize all nutrients to per 100 g, and add an explainability layer that shows which flags drove the score. Do not use a blanket margarine penalty.

## Assessment

This update needs a regression plan because the intended behavior is nuanced: some margarines should go **up**, some shortenings should go **down sharply**, and butter should improve on processing while still losing on saturated fat.

## Owner

## QA & Audit Lead

### Why this owner

This is where Bari can easily fail silently through misclassification or double counting.

### Copy-paste prompt

Create a QA suite for Bari's margarine/shortening update. Include at minimum: legacy PHO pie crust, butter croissant, modern soft canola spread, palm-based cookie shortening, IE frosting fat, and sterol-fortified spread. Validate ingredient parsing, per-100 g normalization, cross-jurisdiction label handling, score precedence, and no-double-counting between core nutrient scoring and the new fat-technology module.

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<https://www.who.int/news-room/fact-sheets/detail/trans-fat>

2 9 16 21 51 <https://pmc.ncbi.nlm.nih.gov/articles/PMC9991743/>  
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